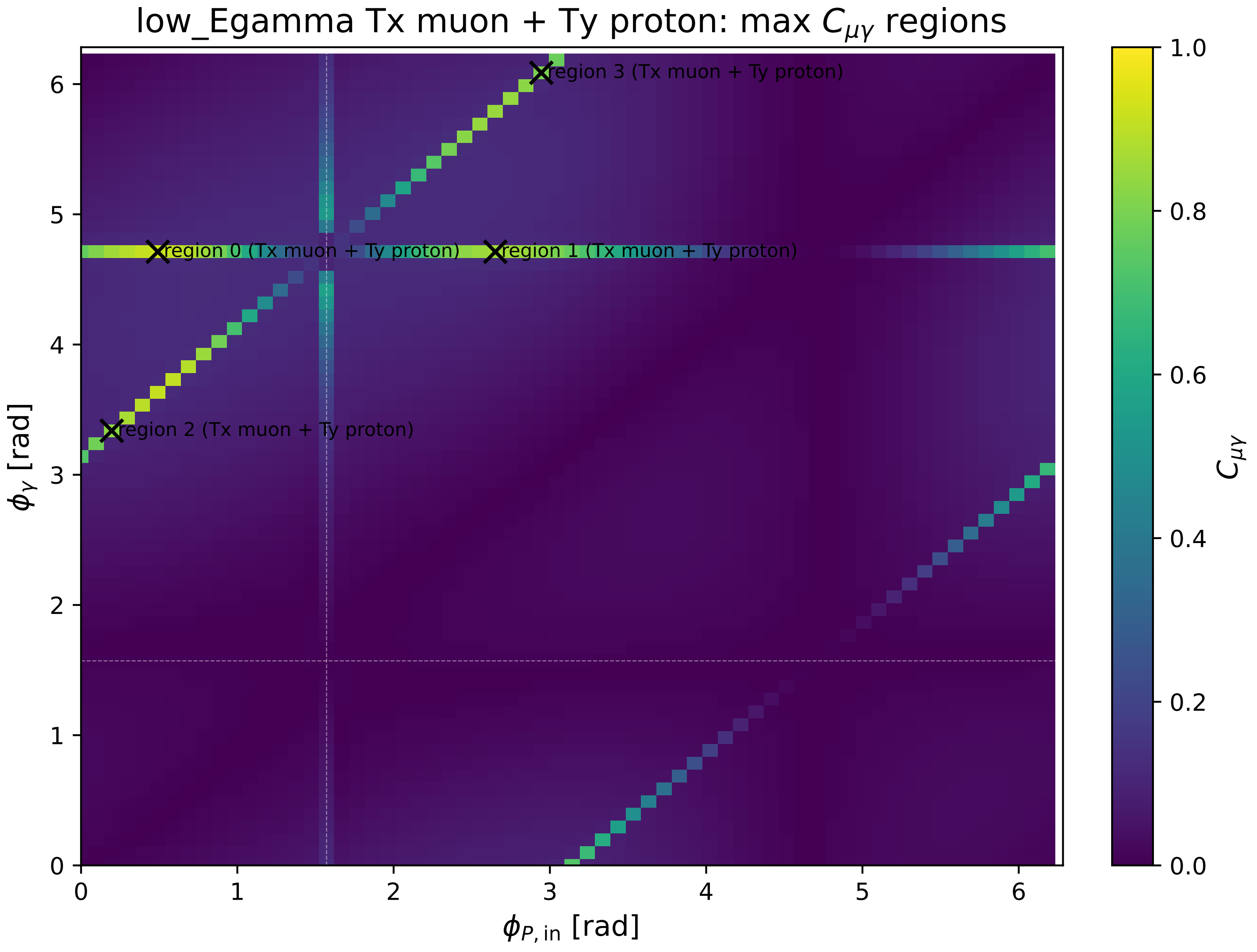
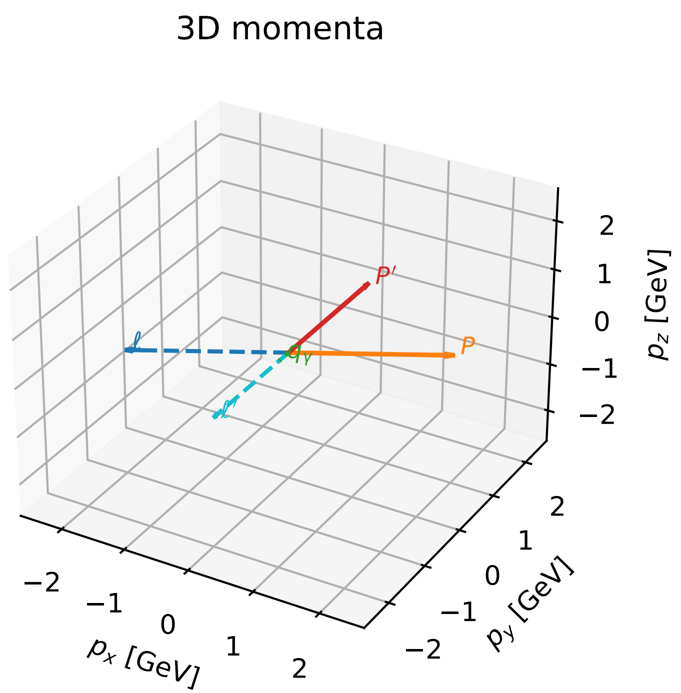

low_Egamma Tx muon + Ty proton: max $C_{\mu\gamma}$ regions
region 0 (Tx muon + Ty proton)
region 1 (Tx muon + Ty proton)
region 2 (Tx muon + Ty proton)
region 3 (Tx muon + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx muon + Ty proton characteristic max $C_{\mu\gamma}$ configuration

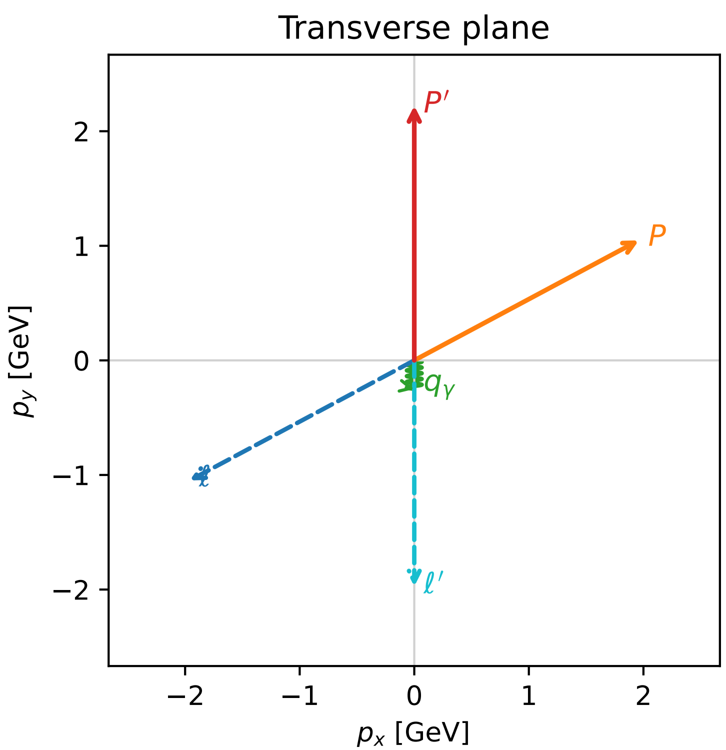


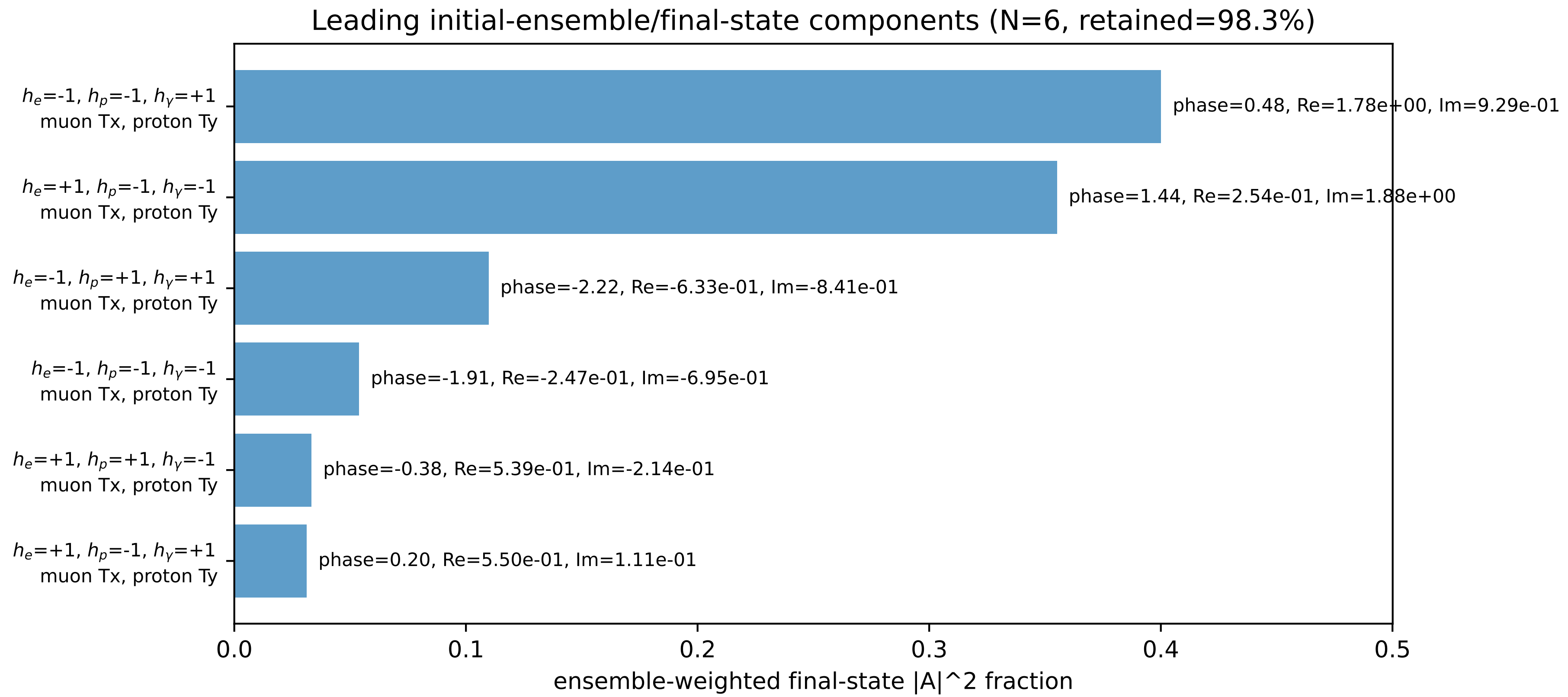
c_e_gamma_low_egamma_txty_region_0 C_{mu\gamma}=0.920122
region: low_Egamma_TxTy_region_0 final-pair delta_xy=0 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.22295
θ_{in}=1.57079, ϕ_ℓ=3.63247, ϕ_P=0.490874
E_γ=0.25, ϕ_γ=4.71239
Q^2=4.63715, x_B=0.666737, t=-5.22456, W^2=3.19769, y=0.337215
$\theta(\ell',\gamma)$=0 deg

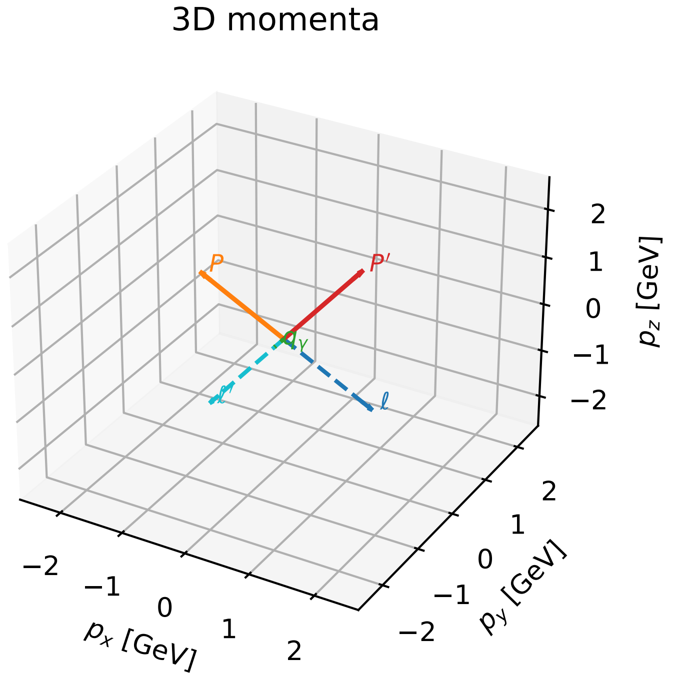
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2256	p_x= -1.9606	p_y= -1.0480	p_z= -0.0000
P	E= 2.4129	p_x= 1.9606	p_y= 1.0480	p_z= 0.0000
ℓ'	E= 1.9758	p_x= 0.0000	p_y= -1.9729	p_z= 0.0000
P'	E= 2.4127	p_x= 0.0000	p_y= 2.2229	p_z= 0.0000
q_γ	E= 0.2500	p_x= -0.0000	p_y= -0.2500	p_z= 0.0000





Tx muon + Ty proton characteristic max $C_{\mu\gamma}$ configuration




```
c_e_gamma_low_egamma_txty_region_1  C_{mu\gamma}=0.85997
region: low_Egamma_TxTy_region_1  final-pair delta_xy=0 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.22295

$\theta_{\rm in}$=1.57079, ϕ_ℓ=5.79231, ϕ_P=2.65072

E_γ=0.25, ϕ_γ=4.71239

Q^2=4.63715, x_B=0.666737, t=-5.22456, W^2=3.19769, y=0.337215

$\theta(\ell',\gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:

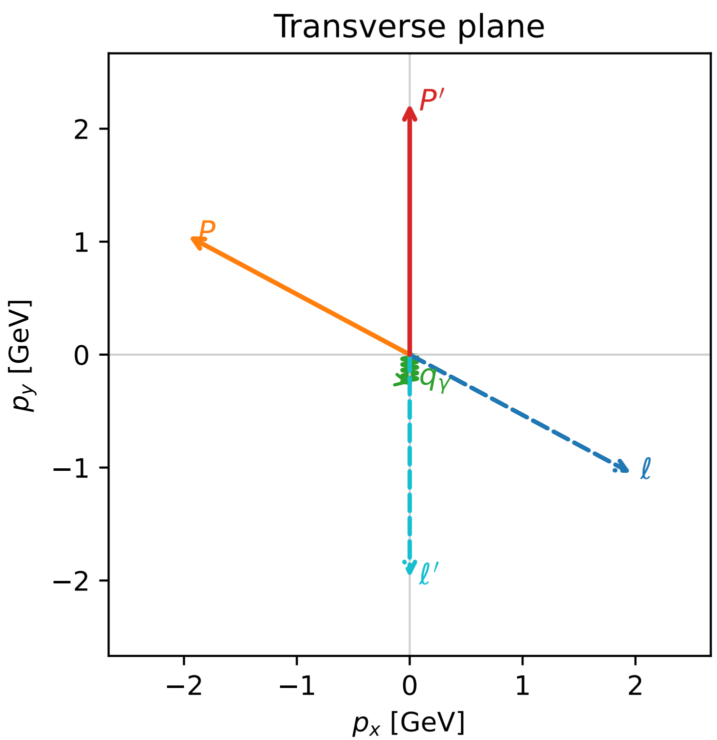
ℓ E= 2.2256 p_x= 1.9606 p_y= -1.0480 p_z= -0.0000

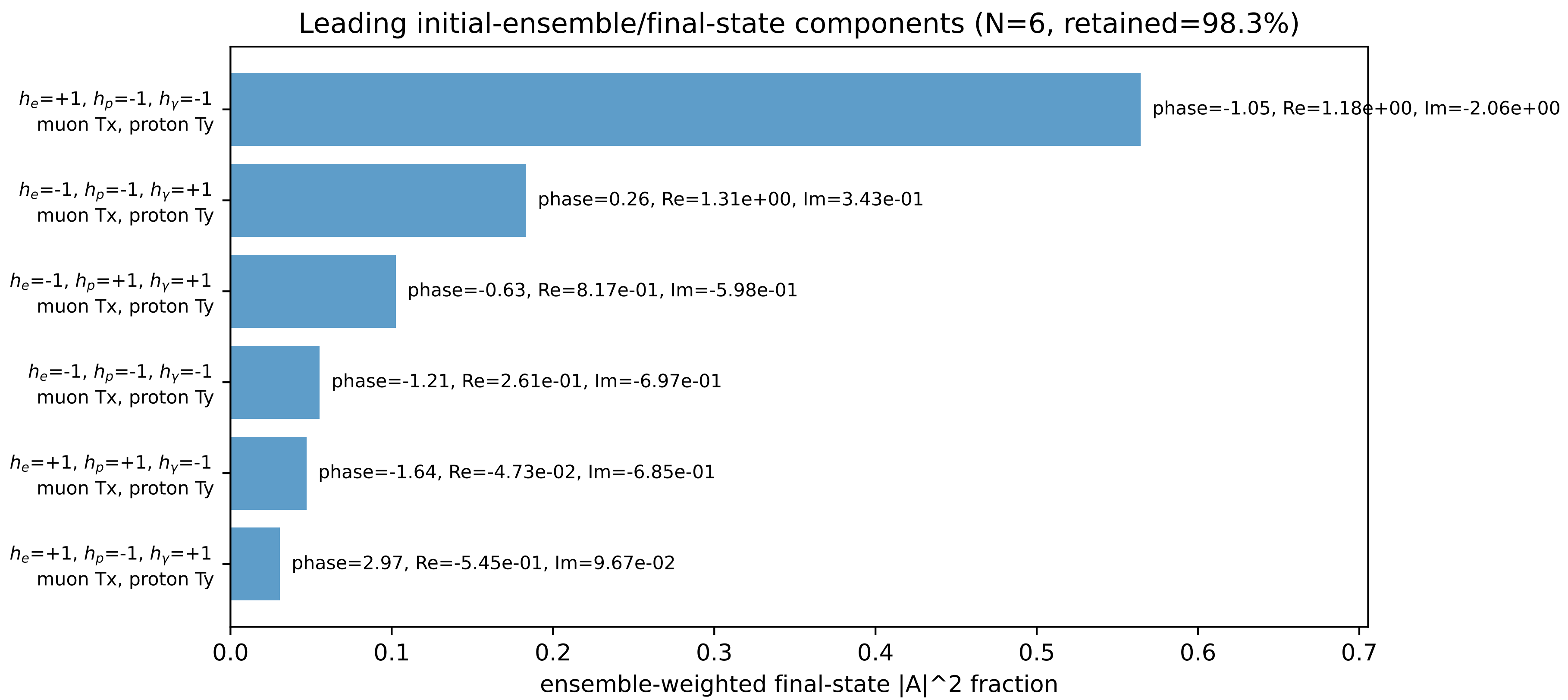
P E= 2.4129 p_x= -1.9606 p_y= 1.0480 p_z= 0.0000

ℓ' E= 1.9758 p_x= 0.0000 p_y= -1.9729 p_z= 0.0000

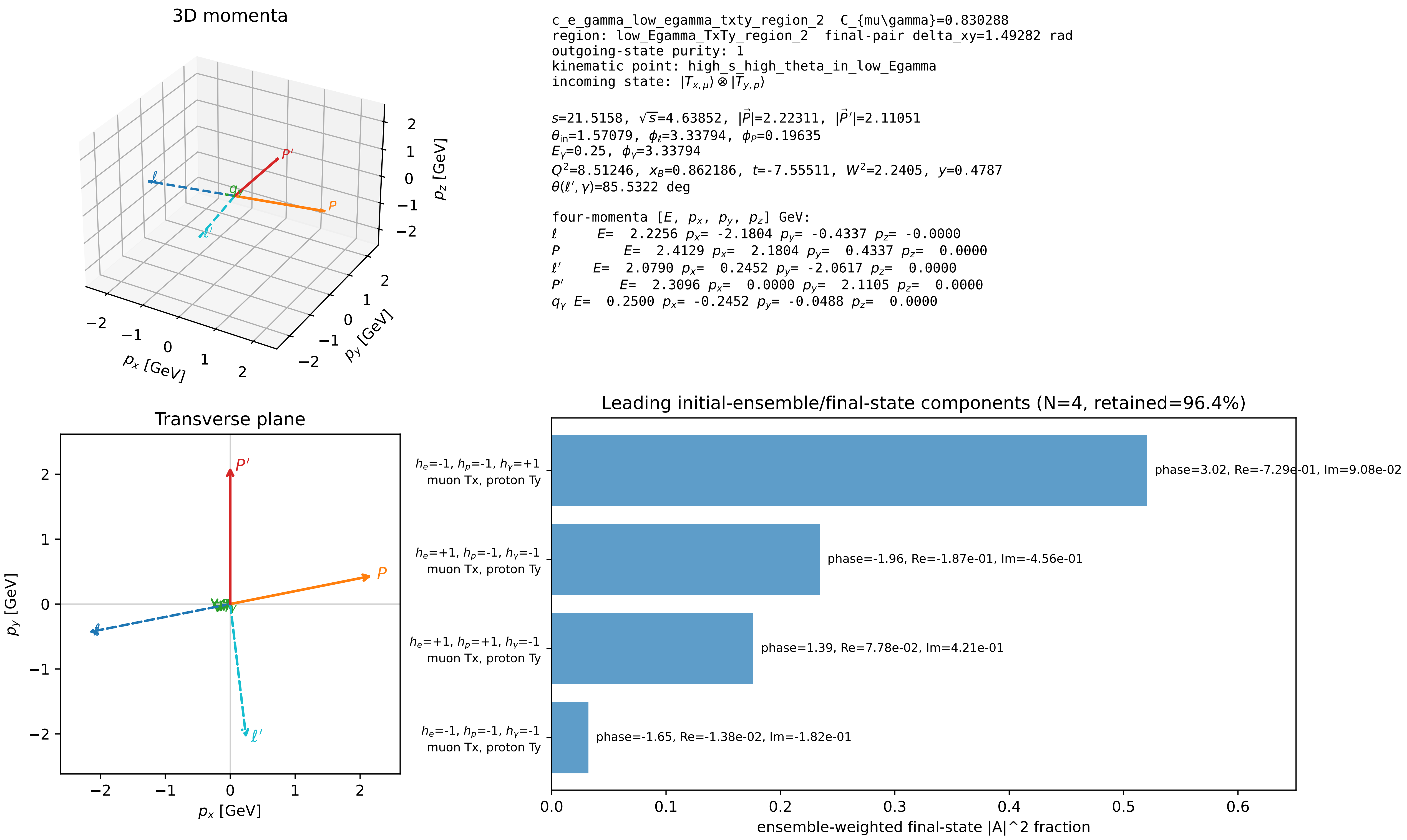
P' E= 2.4127 p_x= 0.0000 p_y= 2.2229 p_z= 0.0000

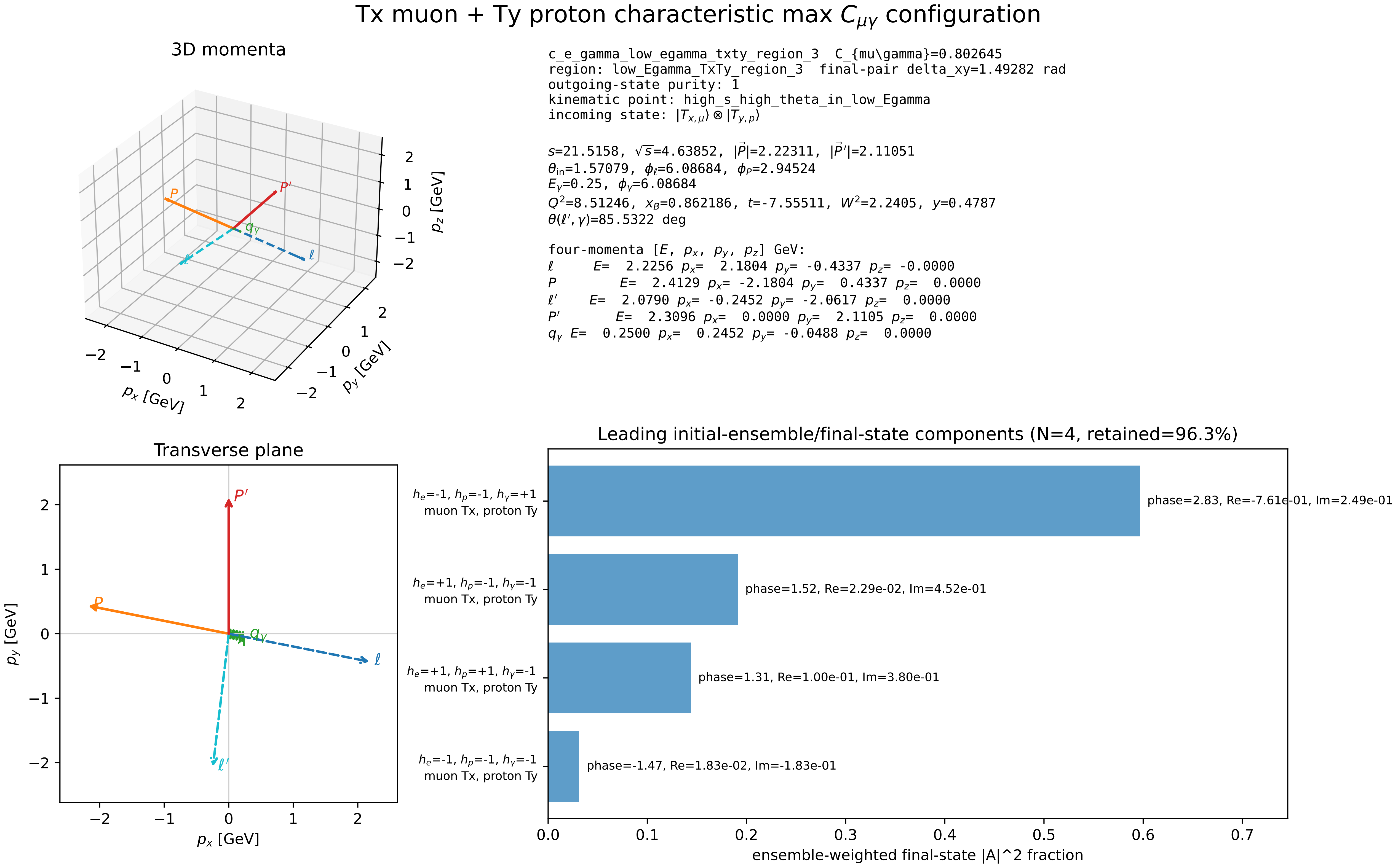
q_γ E= 0.2500 p_x= -0.0000 p_y= -0.2500 p_z= 0.0000





Tx muon + Ty proton characteristic max $C_{\mu\gamma}$ configuration





Tx muon + Ty proton characteristic max $C_{\mu\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ
ℓ'
c_e_gamma_low_egamma_txty_region_3 C_{mu\gamma}=0.802645
region: low_Egamma_TxTy_region_3 final-pair delta_xy=1.49282 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.11051
θ_{in}=1.57079, ϕ_ℓ=6.08684, ϕ_P=2.94524
E_γ=0.25, ϕ_γ=6.08684
Q^2=8.51246, x_B=0.862186, t=-7.55511, W^2=2.2405, y=0.4787
$\theta(\ell', \gamma)$=85.5322 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.1804 p_y= -0.4337 p_z= -0.0000
P E= 2.4129 p_x= -2.1804 p_y= 0.4337 p_z= 0.0000
ℓ' E= 2.0790 p_x= -0.2452 p_y= -2.0617 p_z= 0.0000
P' E= 2.3096 p_x= 0.0000 p_y= 2.1105 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2452 p_y= -0.0488 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=4, retained=96.3%)
h_e=-1, h_p=-1, h_γ=+1 muon Tx, proton Ty
phase=2.83, Re=-7.61e-01, Im=2.49e-01
h_e=+1, h_p=-1, h_γ=-1 muon Tx, proton Ty
phase=1.52, Re=2.29e-02, Im=4.52e-01
h_e=+1, h_p=+1, h_γ=-1 muon Tx, proton Ty
phase=1.31, Re=1.00e-01, Im=3.80e-01
h_e=-1, h_p=-1, h_γ=-1 muon Tx, proton Ty
phase=-1.47, Re=1.83e-02, Im=-1.83e-01
ensemble-weighted final-state |A|^2 fraction